

Cosmetic outcome and seroma formation after breast-conserving surgery with intraoperative radiation therapy boost for early breast cancer.

PMID: 22560552 | Indexed in PubMed

To evaluate cosmetic outcome and its association with breast wound seroma after breast-conserving surgery (BCS) with targeted intraoperative radiation therapy (tIORT) boost for early breast cancer. An analysis of a single-arm prospective study of 55 patients with early breast cancer treated with BCS and tIORT boost followed by conventional whole breast radiation therapy (WBRT) between August 2003 and January 2006 was performed. A seroma was defined as a fluid collection at the primary tumor resection site identified clinically or radiologically. Cosmetic assessments using the European Organization for Research and Treatment of Cancer rating system were performed at baseline before BCS and 30 months after WBRT was completed. Twenty-eight patients (51%) developed a seroma, with 18 patients (33%) requiring at least 1 aspiration. Tumor location was significantly associated with seroma formation ($P=.001$). Ten of 11 patients with an upper inner quadrant tumor developed a seroma. Excellent or good overall cosmetic outcome at 30 months was observed in 34 patients (62%, 95% confidence interval 53%-80%). Seroma formation was not associated with the overall cosmetic result ($P=.54$). BCS with tIORT boost followed by WBRT was associated with an acceptable cosmetic outcome. Seroma formation was not significantly associated with an adverse cosmetic outcome.

Effect of mechanical closure of dead space on seroma formation after breast surgery.

PMID: 16929119 | Indexed in PubMed

Seroma, a collection of serous fluid, is the most common problem after breast surgery. The aim of this study was to review the effect of mechanical closure of dead space on seroma formation after breast surgery. There are two mechanical methods for closure of the dead space beneath skin flaps: compression by external pressure, and fixation of the flaps with sutures. There is no evidence to suggest that the routine use of a pressure garment or compression dressing is beneficial. However, it appears that suture flap fixation does reduce seroma formation, simplify postoperative care and dressing, and thus allow early removal of drains and discharge. Drains are helpful for mechanical closure of dead space, but in breast-conserving surgery this technique may preclude the use of a drain. However, previous studies have often been small in scale and poor in quality. Further studies should examine the effect of this technique on quality of life, cosmesis, and medical economics.

The tumor suppressor MIR139 is silenced by POLR2M to promote AML oncogenesis.

PMID: 34741119 | Indexed in PubMed

MIR139 is a tumor suppressor and is commonly silenced in acute myeloid leukemia (AML). However, the tumor-suppressing activities of miR-139 and molecular mechanisms of MIR139-silencing remain largely unknown. Here, we studied the poorly prognostic MLL-AF9 fusion protein-expressing AML. We show that MLL-AF9 expression in hematopoietic precursors caused epigenetic silencing of MIR139, whereas overexpression of MIR139 inhibited *in vitro* and *in vivo* AML outgrowth. We identified novel miR-139 targets that mediate the tumor-suppressing activities of miR-139 in MLL-AF9 AML. We revealed that two enhancer regions control MIR139 expression and found that the polycomb repressive complex 2 (PRC2)

downstream of MLL-AF9 epigenetically silenced MIR139 in AML. Finally, a genome-wide CRISPR-Cas9 knockout screen revealed RNA Polymerase 2 Subunit M (POLR2M) as a novel MIR139-regulatory factor. Our findings elucidate the molecular control of tumor suppressor MIR139 and reveal a role for POLR2M in the MIR139-silencing mechanism, downstream of MLL-AF9 and PRC2 in AML. In addition, we confirmed these findings in human AML cell lines with different oncogenic aberrations, suggesting that this is a more common oncogenic mechanism in AML. Our results may pave the way for new targeted therapy in AML.

Mechanistic Insight into the Role of Peptides Secreted from *Bacillus clausii* and Future Opportunities.

PMID: 38375835 | Indexed in PubMed

Bacillus clausii is a commercial spore probiotic known to treat multiple diseases. An increased interest in exploring the nutraceutical and probiotic properties of various microorganisms has made researchers explore more about these bacteria. The current trends in the healthcare industry are majorly focused on devising new therapies to avoid drug and pathogen resistance in patients. Antimicrobial peptides have been considered a source of antibiotics for a long time. Still, getting new therapies into the market is a big challenge. Members of the genus *Bacillus* have been reported to have a broad spectrum of antimicrobial peptides. One of the least explored species under this genus is *Bacillus clausii*, concerning peptide drug therapy. The applications of *Bacillus clausii* in treating or preventing gut dysbiosis and respiratory infections have been largely supported in the past two decades. Yet research is lacking in explaining the pathways at molecular levels in targeting pathogens. In this mini-review, we are going to summarise the research that has been reported so far about peptide extraction from *Bacillus clausii*, their mode of action and advantages to mankind, and the challenges lying in the isolation of peptides.

Immunomodulatory and Antioxidant Properties of a Novel Potential Probiotic *Bacillus clausii* CSI08.

PMID: 36838205 | Indexed in PubMed

Spore-forming bacteria of the *Bacillus* genus have demonstrated potential as probiotics for human use. *Bacillus clausii* have been recognized as efficacious and safe agents for preventing and treating diarrhea in children and adults, with pronounced immunomodulatory properties during several in vitro and clinical studies. Herein, we characterize the novel strain of *B. clausii* CSI08 (Munispor®) for probiotic attributes including resistance to gastric acid and bile salts, the ability to suppress the growth of human pathogens, the capacity to assimilate wide range of carbohydrates and to produce potentially beneficial enzymes. Both spores and vegetative cells of this strain were able to adhere to a mucous-producing intestinal cell line and to attenuate the LPS- and Poly I:C-triggered pro-inflammatory cytokine gene expression in HT-29 intestinal cell line. Vegetative cells of *B. clausii* CSI08 were also able to elicit a robust immune response in U937-derived macrophages. Furthermore, *B. clausii* CSI08 demonstrated cytoprotective effects in in vitro cell culture and in vivo *C. elegans* models of oxidative stress. Taken together, these beneficial properties provide strong evidence for *B. clausii* CSI08 as a promising potential probiotic.

Transforming Psoriasis Care: Probiotics and Prebiotics as Novel Therapeutic Approaches.

PMID: 37446403 | Indexed in PubMed

Psoriasis is a chronic inflammatory skin disease with autoimmune pathological characteristics. Recent research has found a link between psoriasis, inflammation, and gut microbiota dysbiosis, and that probiotics and prebiotics provide benefits to patients. This 12-week open-label, single-center clinical trial evaluated the efficacy of probiotics (*Bacillus indicus* (HU36), *Bacillus subtilis* (HU58), *Bacillus coagulans* (SC208), *Bacillus licheniformis* (SL307), and *Bacillus clausii* (SC109)) and precision prebiotics (fructooligosaccharides, xylooligosaccharides, and galactooligosaccharides) in patients with psoriasis receiving topical therapy, with an emphasis on potential metabolic, immunological, and gut microbiota changes. In total, 63 patients were evaluated, with the first 42 enrolled patients assigned to the intervention group and the next 21 assigned to the control group (2:1 ratio; non-randomized). There were between-group differences in several patient characteristics at baseline, including age, psoriasis severity (the incidence of severe psoriasis was greater in the intervention group than in the control group), the presence of nail psoriasis, and psoriatic arthritis, though it is not clear whether or how these differences may have affected the study findings. Patients with psoriasis receiving anti-psoriatic local therapy and probiotic and prebiotic supplementation performed better in measures of disease activity, including Psoriasis Area and Severity Index, Dermatology Life Quality Index, inflammatory markers, and skin thickness compared with those not receiving supplementation. Furthermore, in the 15/42 patients in the intervention group who received gut microbiota analysis, the gut microbiota changed favorably following 12 weeks of probiotic and prebiotic supplementation, with a shift towards an anti-inflammatory profile.

Bacillus clausii probiotic strains: antimicrobial and immunomodulatory activities.

PMID: 15220667 | Indexed in PubMed

The clinical benefits observed with probiotic use are mainly attributed to the antimicrobial substances produced by probiotic strains and to their immunomodulatory effects. Currently, the best-documented probiotic bacteria used in human therapy are lactic acid bacteria. In contrast, studies aiming to characterize the mechanisms responsible for the probiotic beneficial effects of *Bacillus* are rare. The current work seeks to contribute to such characterization by evaluating the antimicrobial and immunomodulatory activities of probiotic *B. clausii* strains. *B. clausii* strains release antimicrobial substances in the medium. Moreover, the release of these antimicrobial substances was observed during stationary growth phase and coincided with sporulation. These substances were active against Gram-positive bacteria, in particular against *Staphylococcus aureus*, *Enterococcus faecium*, and *Clostridium difficile*. The antimicrobial activity was resistant to subtilisin, proteinase K, and chymotrypsin treatment, whereas it was sensitive to pronase treatment. The evaluation of the immunomodulatory properties of probiotic *B. clausii* strains was performed in vitro on Swiss and C57 Bl/6j murine cells. The authors demonstrate that these strains, in their vegetative forms, are able to induce NOS II synthetase activity, IFN-gamma production, and CD4 T-cell proliferation.

Current Progress and Future Perspectives on the Use of Bacillus clausii.

PMID: 35744764 | Indexed in PubMed

Bacillus clausii is a probiotic that benefits human health. Its key characteristics include the ability to form spores; the resulting tolerance to heat, acid, and salt ensures safe passage through the human gastrointestinal tract with no loss of cells. Although *B. clausii* has been widely used for many decades, the beneficial properties of other probiotics, such as *Lactobacillus* spp. and *Bifidobacterium* spp., are better disseminated in the literature. In this review, we summarize the physiological, antimicrobial, and immunomodulatory properties of probiotic *B. clausii* strains. We also describe findings from studies that have investigated *B. clausii* probiotics from the perspective of quality and safety. We highlight innovative properties based on biochemical investigations of non-probiotic strains of *B. clausii*, revealing that *B. clausii* may have further health benefits in other therapeutic areas.

Survival and persistence of *Bacillus clausii* in the human gastrointestinal tract following oral administration as spore-based probiotic formulation.

PMID: 25973914 | Indexed in PubMed

This study aimed to investigate the fate of *Bacillus clausii* spores orally administered as lyophilized or liquid formulation to healthy volunteers. The study was a randomized, open-label, cross-over trial in which two commercial probiotic formulations containing spores of four antibiotic-resistant *B. clausii* strains (OC, NR, SIN, T) were given as a single dose administration. Faecal *B. clausii* units of each strain were counted on selective media and extrapolated for the total weight of evacuated faeces. RAPD-PCR typing was used to confirm *B. clausii* identification. *Bacillus clausii* was found alive in faeces for up to 12 days. In some volunteers, the recovered amount of OC, NR or SIN was higher than the number of administered spores. Bioequivalence among the two formulations was demonstrated. *Bacillus clausii* spores survive transit through the human gastrointestinal tract. They can undergo germination, outgrowth and multiplication as vegetative forms. *Bacillus clausii* strains can have different ability to survive in the intestinal environment. *Bacillus clausii* spores administered as liquid suspension or lyophilized form behave similarly in vivo. This work contributes towards a better understanding of the behaviour of *B. clausii* spores as probiotics.

Gut immune homeostasis: the immunomodulatory role of *Bacillus clausii*, from basic to clinical evidence.

PMID: 35674642 | Indexed in PubMed

The gut microbiota affects the development of the gut immune system in early life. Perturbations to microbiota structure and composition during this period can have long-term consequences on the health of the individual, through its effects on the immune system. Research in the last few decades has shown that probiotic administration can reverse these effects in strain- and environment-specific ways. *Bacillus clausii* (*B. clausii*) has been in use for many decades as a safe and efficacious probiotic, but its mode of action has not yet been completely elucidated. In this review, we discuss how the gut immune system works, the factors that affect its functioning, and the plethora of research highlighting its role in various diseases. We also discuss the known modes of action of *Bacillus* probiotics, and highlight the preclinical and clinical evidence that reveal how *B. clausii* acts to bolster gut defense. We anticipate that the treatment and/or prevention of dysbiosis will be central to managing human health and disease in the future. Discovering the pathophysiology of autoimmune diseases, infections, allergies, and some cancers

will aid our understanding of the key role played by microbial communities in these diseases.
